# 1. Clinical Question & Core Hypothesis

## Clinical Question

Why do organized local immune structures (TLS) sometimes fail to confer a survival advantage? The single-chain path links tumor-suppressor epigenetic loss (RASSF1A) with an immune checkpoint (LAG3), suggesting epigenetic state may reprogram tumor–immune interactions within lymphoid niches. 2) Tier 1 paths used: RASSF1A (tumor-suppressor frequently silenced via promoter methylation) → alteration of tumor-intrinsic programs that can remodel the microenvironment, and LAG3 (immune checkpoint mediating T cell dysfunction). These directly address a mechanism by which tumor epigenetics could convert organized immune infiltrates into ineffective/exhausted responses. 3) Tier 2 supporting evidence: availability in TCGA-BRCA of promoter methylation data (to measure RASSF1A silencing), RNA-seq for LAG3 expression, and H&E WSIs enabling TLS-like aggregate scoring; literature reports that epigenetic silencing can alter antigenicity and immune-modulatory cytokine programs, consistent with a link to checkpoint upregulation/exhaustion. 4) Why OS was chosen: overall survival is the most robust, clinically meaningful endpoint available across TCGA-BRCA and is appropriate to detect a net effect of immune organization modulated by tumor epigenetics. 5) Deliberate exclusions: Non-gene nodes from the path (ultrasound, bladder, urothelial carcinoma, READ/prostate context) were excluded as they are peripheral, cross-tissue observations that do not form a mechanistic backbone for BRCA-focused analysis. No additional genes were included to keep the hypothesis sharply testable in the BRCA dataset?

## Working Hypothesis

In TCGA-BRCA, TLS-high tumors with RASSF1A promoter hypermethylation will exhibit higher LAG3 expression and T cell exhaustion signatures, leading to attenuated antitumor immunity and worse overall survival (OS) compared with TLS-high tumors with unmethylated RASSF1A. The protective association of TLS on survival is conditionally dependent on the epigenetic state of RASSF1A and the functional status of LAG3 signaling; this interaction may vary by intrinsic breast cancer subtype. This hypothesis is testable via integrated analysis of DNA methylation (RASSF1A promoter), LAG3 expression, TLS scoring from histology or transcriptomic proxies, and OS in TCGA-BRCA, with mediation/interaction analyses to evaluate the proposed mechanism.

| Element | Value |
|---|---|
| Imaging feature | TLS-like lymphoid aggregates |
| Anchor gene(s) | RASSF1A, LAG3 |
| Primary endpoint | OS |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**RASSF1A:** RASSF1A silencing), RNA-seq for LAG3 expression, and H&E WSIs enabling TLS-like aggregate scoring; literature reports that epigenetic silencing can alter antigenicity and immune-modulatory cytokine programs, consistent with a link to checkpoint upregulation/exhaustion.

**LAG3:** LAG3 expression, and H&E WSIs enabling TLS-like aggregate scoring; literature reports that epigenetic silencing can alter antigenicity and immune-modulatory cytokine programs, consistent with a link to checkpoint upregulation/exhaustion.

# 3. Clinically Testable Predictions

**Prediction 1:** In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

# 4. Minimum Viable Validation Plan

## A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

## B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

## C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

## Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

## Alternative Directions